

5G RAN

Emergency Communications Solution (Low-Frequency TDD) Guide

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<https://www.huawei.com/en/psirt/vul-response-process>

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue does not introduce any changes to 5G RAN9.1 02 (2025-04-27).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
None	None	4 Coverage Pattern for Emergency and Normal Services

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Coverage pattern for emergency and normal services	None	4 Coverage Pattern for Emergency and Normal Services

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Coverage pattern for emergency and normal services	Supported only in low frequency bands	4 Coverage Pattern for Emergency and Normal Services

3 Overview

Huawei has launched an emergency communications solution to effectively cope with natural disasters, such as earthquakes, winter weather disasters, typhoons, and floods. This solution deals with how to build and use emergency base stations, covering the following aspects:

- **Physical base stations:** To prevent structural collapses and other service-interrupting accidents, it is recommended that earthquake-proof and windproof measures be taken at emergency base stations. These measures help ensure continuous uptime and stable signal coverage in disaster-struck areas. In addition, given the critical importance of signal coverage during emergencies, it is recommended that emergency base stations be placed at higher elevations than normal base stations to expand signal reach.
- **Equipment environment:** To prevent service interruption caused by events such as mains outage, it is recommended that Huawei's various power supply solutions be used at emergency base stations. These solutions can provide stable, reliable power supply even when the mains supply becomes unavailable, thereby ensuring stable signal coverage in disaster-struck areas. For details about these solutions, see "Basic Power Supply Management Functions" in *Power Supply Management*. It is also recommended that Power Supply Redundancy be enabled for emergency base stations. This feature enables power supply units working in redundancy mode to share power load. When a power supply unit becomes faulty, its power load automatically switches to another unit, ensuring continuous uptime. For details about this feature, see "Power Supply Redundancy" in *Base Station Equipment Reliability*.
- **Software functions:** To improve coverage or interference suppression, it is recommended that the functions in [Table 3-1](#) be enabled for emergency base stations.

Table 3-1 Software functions related to the emergency communications solution

Function	Principle	Reference
Coverage pattern for emergency and normal services	This function includes "SSB coverage pattern for emergency and normal services" and "connected mode coverage pattern for emergency and normal services". (SSB is short for synchronization signal and PBCH block.) With this function: • During emergencies, emergency base stations can cover as wide an area as possible. • In normal situations, emergency base stations cause little overshooting interference while providing contiguous coverage with surrounding normal base stations.	4 Coverage Pattern for Emergency and Normal Services
Extended Cell Range	This function improves the access capabilities corresponding to random access preamble formats, further expanding the coverage of emergency base stations.	"Extended Cell Range (TDD)" and "Extended Cell Range of 90 km (TDD)" in <i>Extended Cell Range</i>
Fusion Cell	This function combines two transmission reception points (TRPs) working on the same frequency and bandwidth on two massive multiple-input multiple-output (MIMO) active antenna units (AAUs) into one logical cell. This further expands the coverage of emergency base stations.	"Fusion Cell" in <i>Fusion Cell (Low-Frequency TDD)</i>

4 Coverage Pattern for Emergency and Normal Services

4.1 Principles

"Coverage pattern for emergency and normal services" includes two functions: "SSB coverage pattern for emergency and normal services" and "connected mode coverage pattern for emergency and normal services".

SSB Coverage Pattern for Emergency and Normal Services

This function is designed to enable SSB beams to serve both emergency and normal services. It accomplishes this by adjusting the coverage scope of SSB beams from emergency base stations for emergencies and normal situations respectively:

- During emergencies, SSB beams take the shape of a pyramid (by setting the **NRDUCellTrpBeam.CoverageScenario** parameter to **SCENARIO_33**) to ensure access of UEs at the cell edge. This coverage scenario fully considers the coverage difference between UEs at the cell center and UEs at the cell edge. Pyramid-shaped SSB beams consist of two vertical layers:
 - Upper layer: Beams in this layer feature narrow horizontal beamwidths. They are used to ensure coverage for UEs at the cell edge, while minimizing overlap with beams from neighboring cells to reduce inter-cell interference. SSB power aggregation can be performed for these beams (by setting the **NRDUCellTrpBeam.MaxSsbPwrOffsetExt** parameter to a value ranging from 0 to 3) to further improve cell coverage. The aggregation can significantly improve SSB coverage, allowing more UEs at the cell edge to access the cell.
 - Lower layer: Beams in this layer feature wide horizontal beamwidths. They are used to ensure contiguous coverage for UEs at the cell center.

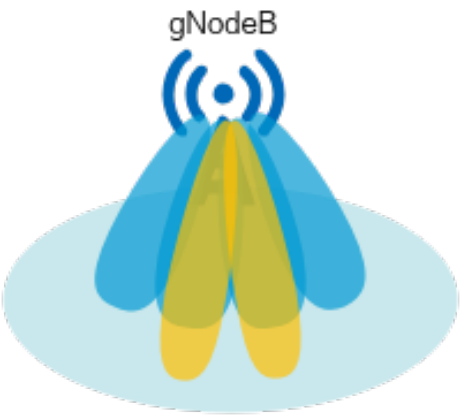
Table 4-1 lists the coverage scenario for emergencies in "SSB coverage pattern for emergency and normal services".

Table 4-1 Coverage scenario for emergencies in "SSB coverage pattern for emergency and normal services"

Coverage Scenario	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_33	65°	12°	0°~9°	-22°~22°

Figure 4-1 shows an example of 2+4 pyramid-shaped beams consisting of two layers. In the upper layer, two beams are directed at 0° vertically, each with a horizontal 3 dB beamwidth of 30° and a vertical 3 dB beamwidth of 6°. In the lower layer, four beams are directed at 6° vertically, each with a horizontal 3 dB beamwidth of 65° and a vertical 3 dB beamwidth of 6°.

Figure 4-1 2+4 pyramid-shaped beams consisting of two layers



- In normal situations, optimized weights are used for SSB beams with a horizontal 3 dB beamwidth of 105° or 65° (by setting the **NRDUCellTrpBeam.CoverageScenario** parameter to **SCENARIO_34** or **SCENARIO_35**, respectively). These coverage scenarios mitigate strong SSB interference from emergency base stations to public-network base stations at lower elevations by suppressing emergency base stations' upper side lobes of SSB beams with large tilt angles. **Table 4-2** lists the coverage scenarios for normal situations in "SSB coverage pattern for emergency and normal services".

Table 4-2 Coverage scenarios for normal situations in "SSB coverage pattern for emergency and normal services"

Coverage Scenario	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_34	105°	6°	-2°~13°	0°

Coverage Scenario	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Tilt Adjustment Range	Azimuth Adjustment Range
SCENARIO_35	65°	6°	-2°~13°	-22°~22°

Connected Mode Coverage Pattern for Emergency and Normal Services

This function is designed to reduce interference between cells served by emergency base stations and neighboring cells in normal situations, while ensuring full coverage during emergencies. It accomplishes this by adjusting the coverage scope of beams from emergency base stations for UEs in connected mode to cater to different scenarios. This function is controlled by the **NRDUCellTrpBeam.ConnModeCovScenarioExt** parameter.

- In normal situations, emergency base stations both cause and suffer from strong interference. In the downlink, high-elevation emergency base stations often cause overshoot coverage due to the absence of obstacles. UEs in cells served by surrounding base stations can therefore receive downlink signals from emergency base stations and experience strong downlink interference. For the same reason, in the uplink, emergency base stations can receive uplink signals from UEs in cells served by surrounding base stations and experience strong interference.

To mitigate inter-cell interference, emergency base stations need to shrink the envelope range of beams for UEs in connected mode to control the range of traffic beams to be selected. This can be achieved by setting the **NRDUCellTrpBeam.ConnModeCovScenarioExt** parameter to a value ranging from **SCENARIO_1** to **SCENARIO_4**. [Table 4-3](#) lists the coverage scope in each scenario.

Table 4-3 Coverage scenarios for normal situations in "connected mode coverage pattern for emergency and normal services"

Coverage Scenario	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Description
SCENARIO_1	105°	6°	Data channels, control channels, and reference signals of a cell cover 105° horizontally and 6° vertically.
SCENARIO_2	105°	12°	Data channels, control channels, and reference signals of a cell cover 105° horizontally and 12° vertically.
SCENARIO_3	65°	6°	Data channels, control channels, and reference signals of a cell cover 65° horizontally and 6° vertically.

Coverage Scenario	Horizontal 3 dB Beamwidth	Vertical 3 dB Beamwidth	Description
SCENARIO_4	65°	12°	Data channels, control channels, and reference signals of a cell cover 65° horizontally and 12° vertically.

- During emergencies, the coverage scope of beams from emergency base stations for UEs in connected mode are maximized (by setting the **NRDUCellTrpBeam.ConnModeCovScenarioExt** parameter to **DEFAULT**). This coverage scenario uses the entire antenna array for beamforming to ensure full coverage of traffic beams.

4.2 Network Analysis

4.2.1 Benefits

Enabling the emergency communications solution for a base station requires the following conditions to be met:

- The base station must be positioned at an elevation exceeding 50 m and must be at least 20 m higher in elevation than any other base station within its coverage area.
- The base station mainly relies on line of sight (LOS) transmission within its cell coverage and is free of obstacles in the main lobe direction.

The functions of the emergency communications solution offer the following benefits:

- SSB coverage pattern for emergency and normal services
 - During emergencies, pyramid-shaped SSB beams expand the coverage of emergency base stations (coverage calculated based on engineering parameters including the elevation and downtilt angle of a base station). In addition, such beams reduce SSB coverage overlap at the cell edge and mitigate SSB interference from emergency base stations to neighboring cells in areas where their coverage overlaps at the cell edge. A larger value of the **NRDUCellTrpBeam.MaxSsbPwrOffsetExt** parameter results in higher power used by the upper-layer beams of pyramid-shaped SSB beams and a longer reach of pyramid-shaped SSB beams. A smaller value of this parameter results in lower power used by the upper-layer beams of pyramid-shaped SSB beams, higher power available for simultaneous data transmission, and given sufficient cell traffic volume, higher **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**.
 - In normal situations, for emergency base stations positioned at higher elevations, this function mitigates SSB interference to neighboring cells in areas where their coverage overlaps at the cell edge.
- Connected mode coverage pattern for emergency and normal services

- In normal situations, for emergency base stations positioned at higher elevations, this function reduces inter-cell interference and increases the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** for both cells served by emergency base stations and cells served by neighboring base stations.
- During emergencies, this function offers no benefits.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- SSB coverage pattern for emergency and normal services
 - During emergencies, this function may change the SSB RSRP and SINR of UEs at the cell edge. As a result, the cells providing the best coverage for these UEs may change, and user experience may change. A larger value of the **NRDUCellTrpBeam.MaxSsbPwrOffsetExt** parameter results in higher power used by the upper-layer beams of pyramid-shaped SSB beams, lower power available for simultaneous data transmission, and given sufficient cell traffic volume, lower **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)**. A smaller value of this parameter results in lower power used by the upper-layer beams of pyramid-shaped SSB beams, and a shorter reach of pyramid-shaped SSB beams.
 - In normal situations, this function may change the SSB RSRP and SINR of UEs at the cell center. As a result, the cells providing the best coverage for these UEs may change, and user experience may change.
- Connected mode coverage pattern for emergency and normal services
 - Assume that a base station mainly employs line of sight (LOS) transmission within its coverage and is free from obstacles and reflections in the beam direction. With a large downtilt angle, there may be many interfering signals on the upper-layer beams in the vertical plane. For such a base station, if the **NRDUCellTrpBeam.ConnModeCovScenarioExt** parameter is set to a value other than **DEFAULT** to shrink the envelope range of beams for UEs in connected mode, the **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** may remain unchanged or slightly increase.
 - Assume that a base station mainly employs non-line-of-sight (NLOS) transmission within its coverage and is subject to a large number of obstacles and reflections in the beam direction. With a large downtilt angle, there may be many multipath wanted signals on the upper-layer beams in the vertical plane. For such a base station, if the **NRDUCellTrpBeam.ConnModeCovScenarioExt** parameter is set to a value other than **DEFAULT** to shrink the envelope range of beams for UEs in connected mode, the **Cell Uplink Average Throughput (DU)** and **Cell Downlink Average Throughput (DU)** decrease, and some UEs may even experience service drops or access exceptions.
 - Adjusting beam patterns for UEs in connected mode affects the radio coverage scope and changes the values of network KPIs. You are advised to set related parameters based on actual situations.

4.3 Requirements

4.3.1 Licenses

None

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
SSB coverage pattern for emergency and normal services	NRDUCellTrpBeam.CoverageScenario	<i>Emergency Communication's Solution (Low-Frequency TDD)</i>	The NRDUCellTrpBeam.MaxSsbPwrOffsetExt parameter can be set to a non-zero value for a cell only if the NRDUCellTrpBeam.CoverageScenario parameter is set to SCENARIO_33 .
Connected Mode Coverage Scenario	NRDUCellTrpBeam.ConnModeCoverageScenario	None	The NRDUCellTrpBeam.ConnModeCovScenarioExt parameter can be set to a value other than DEFAULT for a cell only if the NRDUCellTrpBeam.ConnModeCoverageScenario parameter is set to DEFAULT .

4.3.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
RF channel intelligent shutdown	RF_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	- If RF channel intelligent shutdown is enabled for a cell, the NRDUCellTrpBeam.CoverageScenario parameter cannot be set to SCENARIO_33, SCENARIO_34, or SCENARIO_35 for the cell. - If RF channel intelligent shutdown is enabled for a cell, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT for the cell.
Customized backup power saving policy	NRDUCellPowerSaving.BackupPwrSavingPolicy set to CUSTOMIZED_BACKUP_POLICY	<i>Green Site</i>	- If a cell uses a customized backup power saving policy for low power consumption mode, the NRDUCellTrpBeam.CoverageScenario parameter cannot be set to SCENARIO_33, SCENARIO_34, or SCENARIO_35 for the cell. - If a cell uses a customized backup power saving policy for low power consumption mode, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT for the cell.

Function Name	Function Switch	Reference	Description
Horizontal Coverage Configuration Policy	NRDUCellPdschPrecode.HorizontalCovConfigurationPolicy	None	If the NRDUCellTrpBeam.Con nModeCovScenarioExt parameter is set to a value other than DEFAULT for a cell, Horizontal Coverage Configuration Policy must be set to CONFIG_OFF .

Function Name	Function Switch	Reference	Description
Configuration of the maximum port number used for CSI-RS for CM	NRDUCellCsirs.FR1MaxCellCsirsPortNum	<i>Channel Management</i>	• If the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 32PORT for a cell, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT or SCENARIO_1 for the cell. • If the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 16PORT for a cell, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT, SCENARIO_1, SCENARIO_2, or SCENARIO_3 for the cell. • If the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 16PORT, the NRDUCellCsirs.CsirsCellResourceNum parameter is set to 1_RESOURCE, and the NRDUCellCsirs.CsirsBeamType is set to TYPE0 for a cell, the NRDUCellTrpBeam.ConnModeCovScenarioExt parameter must be set to DEFAULT for the cell.

Function Name	Function Switch	Reference	Description
3D beam avoidance	NRDUCellPdschPrecode.TridBeamAvoid	None	If the NRDUCellTrpBeam.Con nModeCovScenarioExt parameter is set to a value other than DEFAULT for a cell, the NRDUCellPdschPre-code.TridBeamAvoid parameter must be set to 0 for the cell.
3D beam avoidance extension	NRDUCellPdschPrecode.TridBeamAvoidExt	None	If the NRDUCellTrpBeam.Con nModeCovScenarioExt parameter is set to a value other than DEFAULT for a cell, the NRDUCellPdschPre-code.TridBeamAvoidExt parameter must be set to 0 for the cell.

Function Name	Function Switch	Reference	Description
Number of resources for periodic CSI-RS transmission	NRDUCellCsirs.CsirsCellResourceNum	<i>Channel Management</i>	- If the number of resources for periodic CSI-RS transmission is set to 1_RESOURCE, the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 16PORT, and the NRDUCellCsirs.CsirsBeamType parameter is set to TYPE0 for a cell, the NRDUCellTrpBeam.ConnnModeCovScenarioExt parameter must be set to DEFAULT for the cell. - If the number of resources for periodic CSI-RS transmission is set to FD_RESOURCE for a cell, the NRDUCellTrpBeam.ConnnModeCovScenarioExt parameter cannot be set to SCENARIO_3 or SCENARIO_4 for the cell.
CSI-RS Beam Type	NRDUCellCsirs.CsirsBeamType	<i>Beam Management</i>	If the NRDUCellCsirs.CsirsBeamType parameter is set to TYPE0 , the NRDUCellCsirs.FR1MaxCellCsirsPortNum parameter is set to 16PORT , and the NRDUCellCsirs.CsirsCellResourceNum parameter is set to 1_RESOURCE for a cell, then the NRDUCellTrpBeam.ConnnModeCovScenarioExt parameter must be set to DEFAULT for the cell.

Function Name	Function Switch	Reference	Description
CSI-RS beam sweeping for speed turbo UEs	ST_UE_CSIRS_BEAM_SWEEPING_SW option of the NRDUCellServExp.SpdTurboAlgoSwitch parameter	None	CSI-RS beam sweeping for speed turbo UEs can be enabled in a cell only if the NRDUCellTrpBeam.ConanModeCovScenarioExt parameter is set to DEFAULT for the cell.

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

Boards

All NR-capable main control boards and NR TDD-capable baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function requires 64T64R NR TDD-capable RF modules that support 2.6 GHz. The following RF modules can be used: AAU5614, AAU5636, AAU5636w, and AAU5639w. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Cells

The following requirements must be met for "SSB coverage pattern for emergency and normal services" and "connected mode coverage pattern for emergency and normal services":

- The **NRDUCell.FrequencyBand** parameter must be set to **N41**.
- The **NRDUCellTrp.TxRxMode** parameter must be set to **64T64R**.

4.3.5 Others

None

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-4 describes the parameters used for activating "coverage pattern for emergency and normal services". This section does not describe parameters related to cell establishment.

Table 4-4 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Coverage Scenario	NRDUCellTrpBeam.CoverageScenario	None	Set this parameter based on the network plan. Set this parameter to SCENARIO_33 for "SSB coverage pattern for emergency and normal services" during emergencies, and to SCENARIO_34 or SCENARIO_35 for this function in normal situations.
Max SSB Power Offset Extension	NRDUCellTrpBeam.MaxSsbPwrOffsetExt	None	You are advised to set this parameter based on the network plan for "SSB coverage pattern for emergency and normal services" during emergencies.
Connected Mode Coverage Scenario Extension	NRDUCellTrpBeam.ConnModeCoverageScenarioExt	None	Set this parameter based on the network plan. Set this parameter to a value ranging from SCENARIO_1 to SCENARIO_4 for "connected mode coverage pattern for emergency and normal services" in normal situations, and to DEFAULT for this function during emergencies.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- During emergencies
//Configuring "SSB coverage pattern for emergency and normal services" (using SCENARIO_33)
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=0, CoverageScenario=SCENARIO_33;
//Setting 3 dB power aggregation for the upper-layer beams of pyramid-shaped SSB beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=0, MaxSsbPwrOffsetExt=3;
- In normal situations
//Configuring "SSB coverage pattern for emergency and normal services" (using SCENARIO_34 as an example)
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=0, CoverageScenario=SCENARIO_34;
//Configuring "connected mode coverage pattern for emergency and normal services" (using SCENARIO_2 as an example)
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=0, ConnModeCovScenarioExt=SCENARIO_2;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- During emergencies
//Configuring default SSB beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=0, CoverageScenario=DEFAULT;
//Canceling power aggregation for the upper-layer beams of pyramid-shaped SSB beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=0, MaxSsbPwrOffsetExt=0;
- In normal situations
//Configuring default SSB beams
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=0, CoverageScenario=DEFAULT;
//Configuring default beams for UEs in connected mode
MOD NRDUCELLTRPBEAM: NrDuCellTrpld=0, ConnModeCovScenarioExt=DEFAULT;

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

After this function is enabled, run the **DSP NRDUCELLTRP** command. This function has taken effect if the value of **Actual Beam Scenario** is the configured

scenario for SSB beams, or the value of **Connected Mode Actual Beam Scenario Extension** is the configured scenario for beams for UEs in connected mode.

4.4.3 Network Monitoring

The performance indicators described in [4.2.2 Impacts](#) can be used for network monitoring of this function.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Extended Cell Range*
- *Fusion Cell (Low-Frequency TDD)*
- *Emergency Communications Solution (Low-Frequency TDD)*
- *Energy Conservation and Emission Reduction*
- *Green Site*
- *Channel Management*
- *Beam Management*
- *Power Supply Management*
- *Base Station Equipment Reliability*